

CloudFront Origin Group – Scenario 1: EC2 and S3 Failover

To Begin with the Lab

Summary of the Lab

In this lab, a **CloudFront Origin Group** was created to provide failover between an **Amazon EC2** web server and an **Amazon S3** static website. The primary origin was configured on EC2, and the secondary origin was configured on S3 so CloudFront could switch automatically when the primary server became unavailable. A static backup website was hosted in S3 and made public through a bucket policy and static website hosting configuration. A CloudFront distribution was then created using the S3 website endpoint and the EC2 instance as multiple origins, and the default cache behavior was updated to use the origin group. Finally, the lab verified that when the EC2 origin failed, CloudFront served the S3 backup page instead of showing an error.

Understanding CloudFront Origin Groups

A **CloudFront Origin Group** is a failover setup that keeps a website available by using more than one origin. CloudFront sends traffic to the primary origin first, and if that origin becomes unhealthy or stops responding, it automatically switches to the secondary origin. This exists so users do not see a broken site when a server fails. It is useful when the backup origin can show a maintenance page or a simplified version of the website until the main origin returns.

Example:

An online store uses an **Amazon EC2** instance as the main website and an **Amazon S3** bucket as the backup site. If EC2 goes down, CloudFront serves the S3 page that says the site is temporarily unavailable. This keeps the website reachable and gives users a clear message instead of an error page.

Lab Steps

1. Configure Main Instance

- Sign in to the AWS Management Console.
- Create an **Amazon EC2** instance and configure name as **web-server** and enable **Allow HTTP traffic from the internet**.

EC2 > Instances > Launch an instance

Name and tags Info

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search:

Recents | My AMIs | **Quick Start**

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0db56f446d44f2f09 (64-bit (x86), uefi-preferred) / ami-07c18f602ca3306b9 (64-bit (Arm), uefi)

Summary

Number of instances: Info

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier

[Cancel](#) [Launch instance](#) [Preview code](#)

- Now we will configure it as a web server so we first have to access the instance.
- For this, we will use **Command Prompt** and run the following commands.
- Now type: `cd {location_of_your_security_key}`.
- Now type: `ssh -i {security_key_name} ec2-user@{public_ipv4_address_of_ec2_instance}`.

```
ec2-user@ip-172-31-5-100:~
C:\Users\Shashank>cd downloads
C:\Users\Shashank\Downloads>ssh -i June-2026.pem ec2-user@65.2.73.108
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-5-100 ~]$
```

- Now login as root user by typing this command: **sudo -i**.
- Use **yum install httpd** command to configure and place webpage in Apache on the EC2 instance.

```
root@ip-172-31-6-87:~  
[ec2-user@ip-172-31-6-87 ~]$ sudo -i  
[root@ip-172-31-6-87 ~]# yum install httpd  
Amazon Linux 2023 Kernel Livepatch Repository 392 kB/s | 47 kB 00:00  
Last metadata expiration check: 0:00:01 ago on Fri Jun 12 09:07:49 2026.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.67-1.amzn2023.0.1	amazonlinux	46 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	97 k
apr-util-ldb	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	13 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.67-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.67-1.amzn2023.0.1	amazonlinux	12 k
httpd-tools	x86_64	2.4.67-1.amzn2023.0.1	amazonlinux	80 k
libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k
mailcap	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
Installing weak dependencies:				
apr-util-openssl	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	15 k
mod_http2	x86_64	2.0.39-1.amzn2023.0.1	amazonlinux	167 k
mod_lua	x86_64	2.4.67-1.amzn2023.0.1	amazonlinux	59 k

```
Transaction Summary  
=====
```

Transaction Summary	
Install	13 Packages

```
=====
```

Transaction Summary	
Total download size:	2.4 M

- Replace the default web page with the provided index.html content with the help of vi editor.

```
Installed:  
apr-1.7.5-1.amzn2023.0.4.x86_64  
apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64  
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  
httpd-core-2.4.67-1.amzn2023.0.1.x86_64  
httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch  
httpd-tools-2.4.67-1.amzn2023.0.1.x86_64  
libbrotli-1.0.9-4.amzn2023.0.2.x86_64  
mailcap-2.1.49-3.amzn2023.0.3.noarch  
mod_lua-2.4.67-1.amzn2023.0.1.x86_64  
apr-util-1.6.3-1.amzn2023.0.2.x86_64  
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  
httpd-2.4.67-1.amzn2023.0.1.x86_64  
mod_http2-2.0.39-1.amzn2023.0.1.x86_64  
Completed  
[root@ip-172-31-5-100 ~]# cd /var/www/html/  
[root@ip-172-31-5-100 html]# vi index.html
```

- Use the above commands to change content of the default index.html and paste your code like this or use the code below.

```
root@ip-172-31-5-100:/var/www x + v
<!DOCTYPE html>
<html>
<head>
  <title>CloudFront Origin Group Demo</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #232F3E;
      color: white;
      text-align: center;
      padding-top: 100px;
    }
    .container {
      width: 70%;
      margin: auto;
    }
    h1 {
      color: #FF9900;
    }
    .status {
      font-size: 24px;
      color: #00FF7F;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>CloudFront Origin Group Lab</h1>
    <h2>Primary Origin - Amazon EC2</h2>
    <p class="status">Website is being served from the EC2 Web Server.</p>
    <p>This page demonstrates CloudFront Origin Group failover.</p>
    <p>If this server becomes unavailable, CloudFront will automatically redirect traffic to the S3 backup website.</p>
  </div>
</body>
</html>
..
..
..
..
35,7 ALL
```

```
<!DOCTYPE html>
<html>
<head>
  <title>CloudFront Origin Group Demo</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #232F3E;
      color: white;
      text-align: center;
      padding-top: 100px;
    }
    .container {
      width: 70%;
      margin: auto;
    }
    h1 {
      color: #FF9900;
    }
    .status {
      font-size: 24px;
      color: #00FF7F;
```

```

    }
</style>
</head>
<body>
  <div class="container">
    <h1>CloudFront Origin Group Lab</h1>
    <h2>Primary Origin - Amazon EC2</h2>
    <p class="status">Website is being served from the EC2 Web Server.</p>
    <p>This page demonstrates CloudFront Origin Group failover.</p>
    <p>If this server becomes unavailable, CloudFront will automatically redirect traffic to the
S3 backup website.</p>
  </div>
</body>
</html>

```

- Now enable and run the web server services using these commands

```

systemctl start httpd
systemctl enable httpd

```

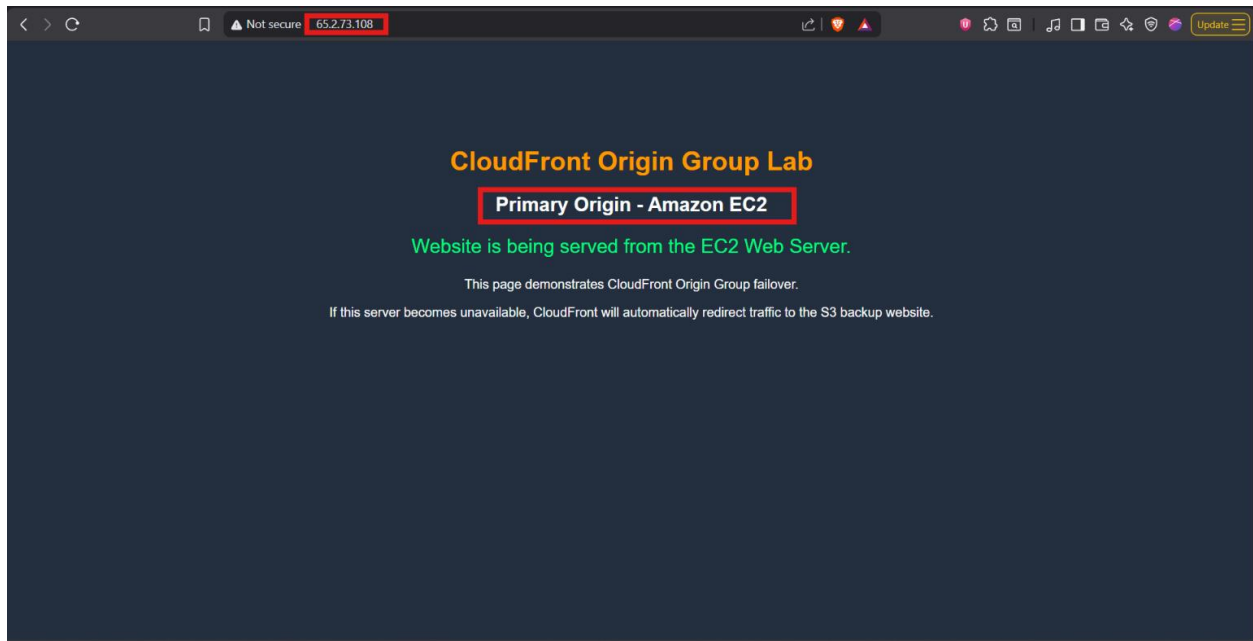
```

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-core-2.4.67-1.amzn2023.0.1.x86_64
httpd-tools-2.4.67-1.amzn2023.0.1.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_lua-2.4.67-1.amzn2023.0.1.x86_64
apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
httpd-2.4.67-1.amzn2023.0.1.x86_64
httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mod_http2-2.0.39-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-5-100 ~]# cd /var/www/html/
[root@ip-172-31-5-100 html]# vi index.html
[root@ip-172-31-5-100 html]# systemctl start httpd
[root@ip-172-31-5-100 html]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-5-100 html]#

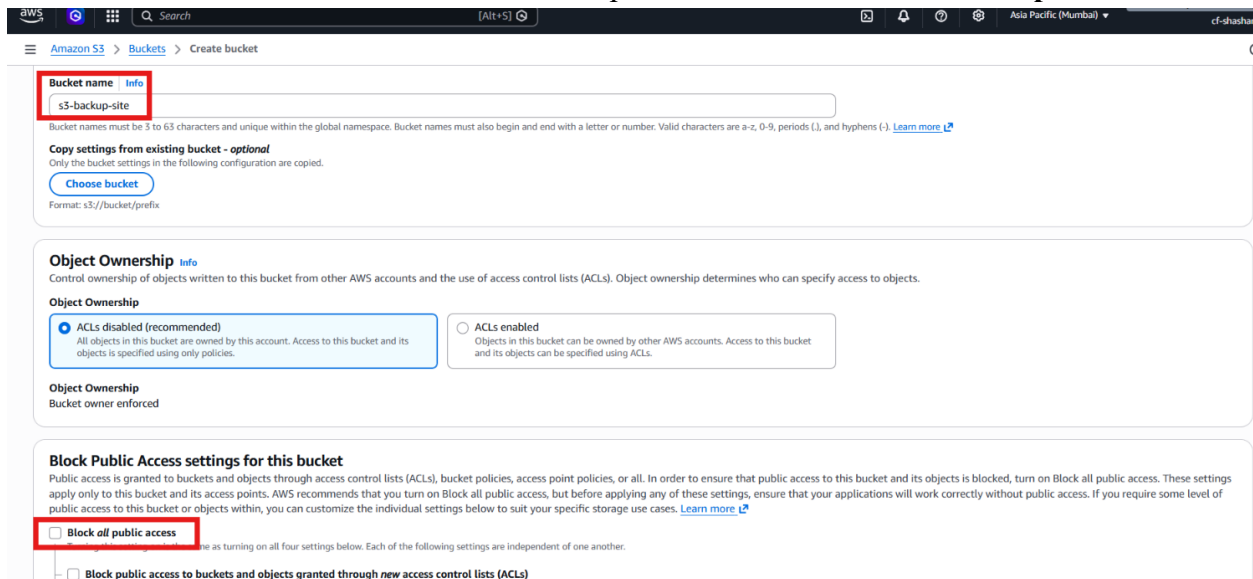
```

- Verify that the EC2 website opens correctly using the public IP.

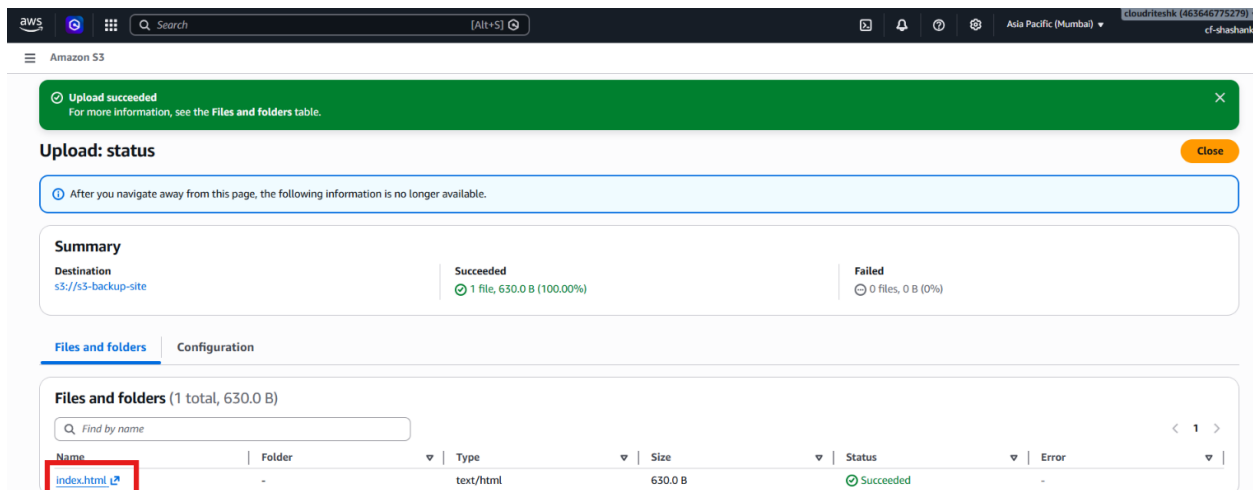


2. Configure Backup Instance

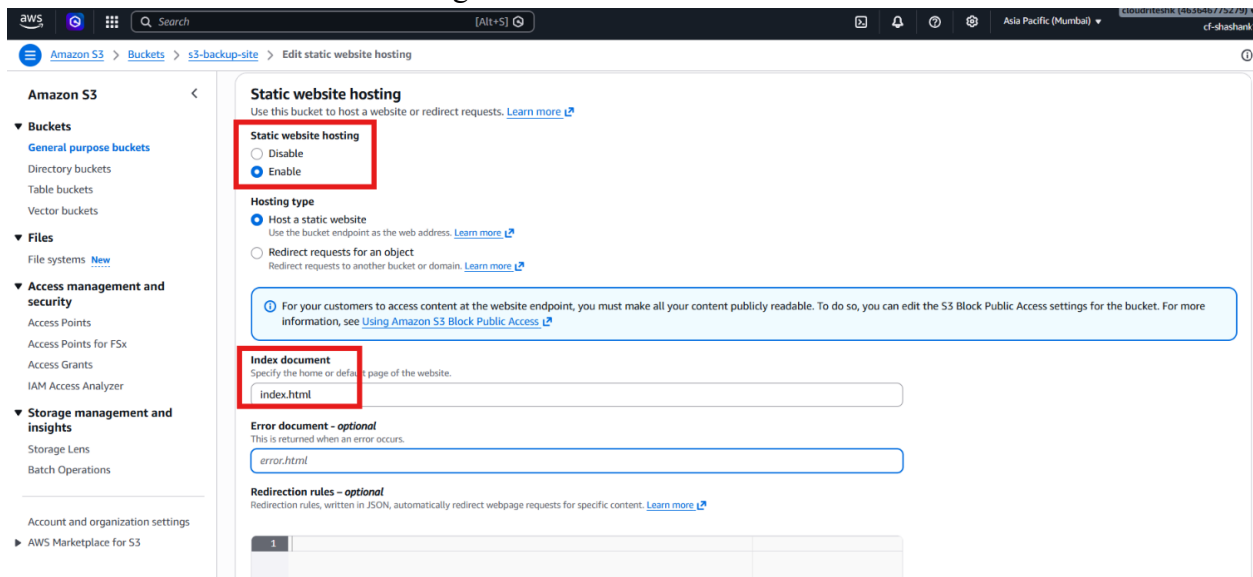
- Create an **Amazon S3** bucket for the backup website and deselect **Block *all* public access**.



- Upload the backup website file, including index.html, to the bucket.



- Make the S3 content public by going to **Properties**, scroll down and enable **Static web hosting**.
- Enable S3 static website hosting and set index.html as the index document.



- Make it public by going to **Permissions** → **Bucket Policy**.
- Click **Edit**.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
```

```
"Resource": "arn:aws:s3:::s3-backup-site/*"
}
]
}
```

Amazon S3 > Buckets > s3-backup-site > Edit bucket policy

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems **New**
- ▼ Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- ▼ Storage management and insights
 - Storage Lens
 - Batch Operations
- Account and organization settings
- AWS Marketplace for S3

Edit bucket policy Info

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN
arn:aws:s3:::s3-backup-site

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "PublicReadGetObject",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "s3:GetObject",
9       "Resource": "arn:aws:s3:::s3-backup-site/*"
10    }
11  ]
12 }
13
```

Edit statement

Select a statement

Select an existing statement in the policy
add a new statement.

[+ Add new statement](#)

- Verify that the S3 website endpoint opens correctly in a browser.

< > ↻

Not secure s3-backup-site.s3-website.ap-south-1.amazonaws.com

Website Temporarily Unavailable

The primary web server is currently unavailable.

You are viewing the backup website hosted on Amazon S3.

Please check back shortly.

3. Configure CloudFront Distribution

- Create a **Amazon CloudFront** distribution.
- Set any name and Choose **Pay as you go** and click **Next**.

Distribution options info

Distribution name
Name will be stored as a tag on the resource. You can change the name, or more tags, later.

cloudfreeks-hosting

Description - optional

Distribution type

☒ **Single website or app**
Choose if each website or application will have a unique configuration.

☐ **Multi-tenant architecture - New**
Choose when you have multiple domains that need to share configurations. This is a common architecture for SaaS providers.

Domain info

Route 53 managed domain - optional
Enter a domain that's already registered with Route 53 in your AWS account. CloudFront will provision a TLS certificate for you. If you have a domain from a different DNS provider, skip this step and configure your domain later.

Check domain

- Select the S3 website endpoint as the first origin and use the website endpoint option.

Origin type

☒ **Amazon S3**
Deliver static content like files and images, statically generated websites or single page applications (SPA).

☐ **Elastic Load Balancer**
Deliver applications hosted behind ELB such as dynamic websites, web services, and APIs.

☐ **API Gateway**
Deliver API endpoints for REST APIs hosted on API Gateway.

☐ **Elemental MediaPackage**
Deliver end-to-end live events or video on demand (VOD).

☐ **VPC origin**
Deliver applications and content hosted within private VPCs, such as EC2 instances and Application Load Balancers.

☐ **Other**
Refer to any AWS or non-AWS origin through its publicly resolvable URL.

Origin

S3 origin
Choose an AWS origin, or enter your origin's domain name. [Learn more](#)

s3-backup-site.s3.ap-south-1.amazonaws.com [Browse S3](#)

Warning: This S3 bucket has static web hosting enabled. If you plan to use this distribution as a website, we recommend using the S3 website endpoint rather than the bucket endpoint. [Use website endpoint](#)

Origin path - optional
The directory path within your origin where your content is stored. [Learn more](#)

/path

- Now go to **Settings** and configure **Cache Settings** as shown for lab purpose to test things faster.

CloudFront > Distributions > Create distribution

Settings Info
CloudFront provides default origin and cache settings based on what origin you selected. [View default settings for S3](#)

Origin settings
Origin settings control how CloudFront connects to the specified origin.

☒ Use recommended origin settings ☐ Customize origin settings

Cache settings
Cache settings determine when CloudFront serves cached content and when it fetches new content from the origin.

☐ Use recommended cache settings tailored to serving S3 content ☒ Customize cache settings

Viewer protocol policy
☐ HTTP and HTTPS
☒ Redirect HTTP to HTTPS
☐ HTTPS only

Allowed HTTP methods
☒ GET, HEAD
☐ GET, HEAD, OPTIONS
☐ GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE

Cache policy
Choose an existing cache policy or create a new one.

Policy with caching disabled

- Choose **Do not enable security protections** for now and create the distribution.

CloudFront > Distributions > Create distribution

Step 1: Choose a plan
Step 2: Get started
Step 3: Specify origin
Step 4: **Enable security**
Step 5: Review and create

Enable security Info

Web Application Firewall (WAF) Info

☐ Enable security protections
Keep your application secure from the most common web threats and security vulnerabilities using AWS WAF. Blocked requests are stopped before they reach your web servers.

☒ **Do not enable security protections.**
Select this option if your application does not need security protections from AWS WAF.

- Now go to **Origins** and click **Create origin**.

CloudFront > Distributions > E3GY39ZK4KGUFC

cloudfreeks-hosting Standard

Details

Distribution domain name: d2s9xlg2s8f0jv.cloudfront.net
Billing: Pay-as-you-go
ARN: arn:aws:cloudfront:463646775:279:distribution/E3GY39ZK4KGUFC
Last modified: Deploying

General | Security | **Origins** | Behaviors | Error pages | Invalidations | Logging | Tags

Origins (1)

Origin name	Origin domain	Origin path	Origin type	Origin Shield reg...	Origin access
☐ s3-backup-site.s3.ap-sout	s3-backup-site.s3-w...		S3 static website	-	-

Origin groups (0)

- Copy the **Public IPv4 DNS** address from **Instance**.

The screenshot shows the AWS Management Console for the 'Instances' page. The 'web-server' instance is selected and highlighted with a red box. Below the instance list, the details for 'i-07f0cb002cf00fbbe (web-server)' are shown. The 'Private IPv4 addresses' section is highlighted with a red box, showing the address 'ec2-65-2-73-108.ap-south-1.compute.amazonaws.com'.

- Paste it as the **Origin domain** and choose **HTTP only** and create the Origin.

The screenshot shows the AWS Management Console for the 'Create origin' page. The 'Origin domain' field is highlighted with a red box and contains the address 'ec2-65-2-73-108.ap-south-1.compute.amazonaws.com'. The 'Protocol' section is also highlighted with a red box, showing 'HTTP only' selected.

- Create an origin group and set EC2 as the primary origin.
- Set the S3 origin as the secondary origin and select all Failover Criteria.

aws CloudFront Distributions E3GY39ZK4KGUFC Create origin group

Origins
Choose the origins for this group, then put them in priority order.

Choose origins to add to group

1: ec2-65-2-73-108.ap-south-1.compute.amazonaws.com (primary) X ▲ ▼

2: s3-backup-site.s3.ap-south-1.amazonaws.com-mq99mep064z X ▲ ▼

Name
Enter a name for this origin group.

ec2-s3

Failover criteria
Select the origin errors to use as failover criteria.

- ☒ 400 Bad Request
- ☒ 403 Forbidden
- ☒ 404 Not found
- ☒ 416 Range Not Satisfiable
- ☒ 429 Too Many Requests
- ☒ 500 Internal server error
- ☒ 502 Bad gateway
- ☒ 503 Service unavailable
- ☒ 504 Gateway timeout

Origin selection criteria
Choose how the origins in this group are selected.

☒ Default

- Go to Behaviors and Edit the default behavior.

cloudfreeks-hosting Standard View metrics

Details

Distribution domain name
d2s9xlg2s8f0jv.cloudfront.net

Billing
Pay-as-you-go
Switch to a plan

ARN
arn:aws:cloudfront:463646775279:distribution/E3GY39ZK4KGUFC

Last modified
June 11, 2026 at 9:14:56 AM UTC

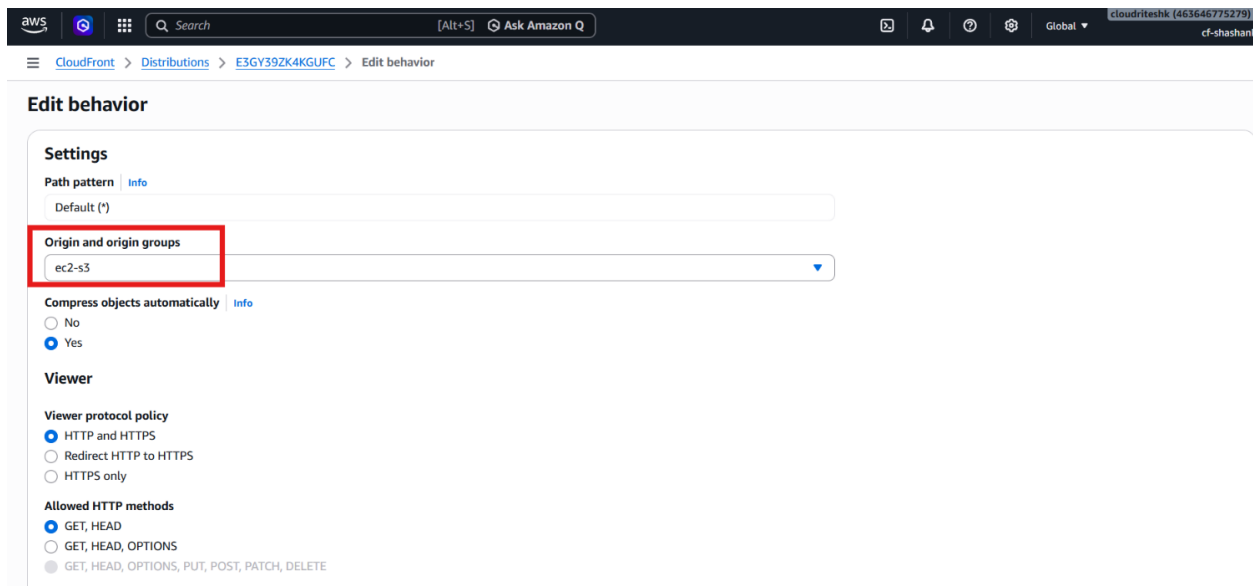
General Security Origins **Behaviors** Error pages Invalidations Logging Tags

Behaviors (1/1) Save Move up Move down Edit Delete Create behavior

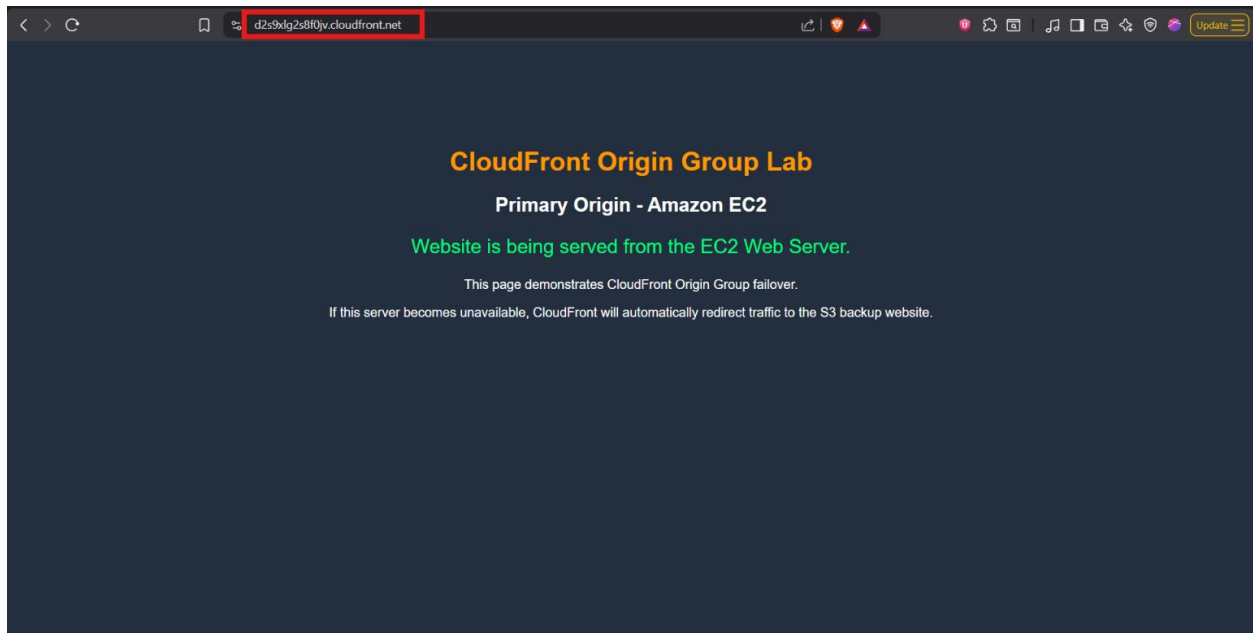
Filter behaviors by property or value

	Preced...	Path pattern	Origin or origin group	Viewer protocol policy	Cache policy name	Origin request policy na...	Response headers policy...
0	Default (*)		s3-backup-site.s3.ap-sout...	Redirect HTTP to HTTPS	Managed-CachingDisabled	-	-

- Now Select the origin group we have created.



- Open the CloudFront distribution URL and verify that the EC2 website loads first.



4. Verification

- Remove the HTTP inbound rule from the EC2 security group to simulate failure.
- For this, select instance and go to **Security**, then select the **Security Group**.

Instances (1/2)

Save filter sets
Choose filter set

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elas
Jenkins-New	i-0e865c7dfb7bf4c89	Stopped	t3.medium	-	ap-south-1b	-	-	-
web-server	i-07f0cb002cf00fbbe	Running	t3.micro	Initializing	ap-south-1b	ec2-65-2-73-108.ap-so...	65.2.73.108	-

i-07f0cb002cf00fbbe (web-server)

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

Security details

IAM role: -
Owner ID: 463646775279
Launch time: Thu Jun 11 2026 13:23:01 GMT+0530 (India Standard Time)

Security groups

sg-0622eddf636d61963 (launch-wizard-1082)

Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups
------	------------------------	------------	----------	--------	-----------------

- **Open Security Group and click Edit Inbound Rule.**

sg-0622eddf636d61963 - launch-wizard-1082

Details

Security group name: launch-wizard-1082
Security group ID: sg-0622eddf636d61963
Description: launch-wizard-1082 created 2026-06-11T07:45:46.876Z
VPC ID: vpc-072c4acd761c3b942

Owner: 463646775279
Inbound rules count: 2 Permission entries
Outbound rules count: 1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

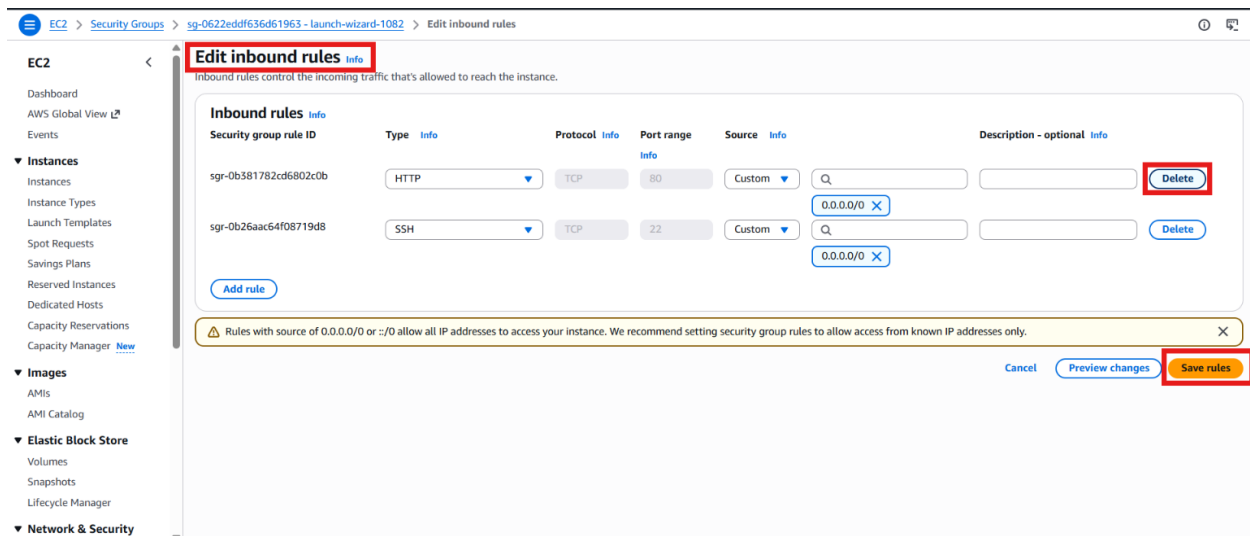
Inbound rules (2)

Search

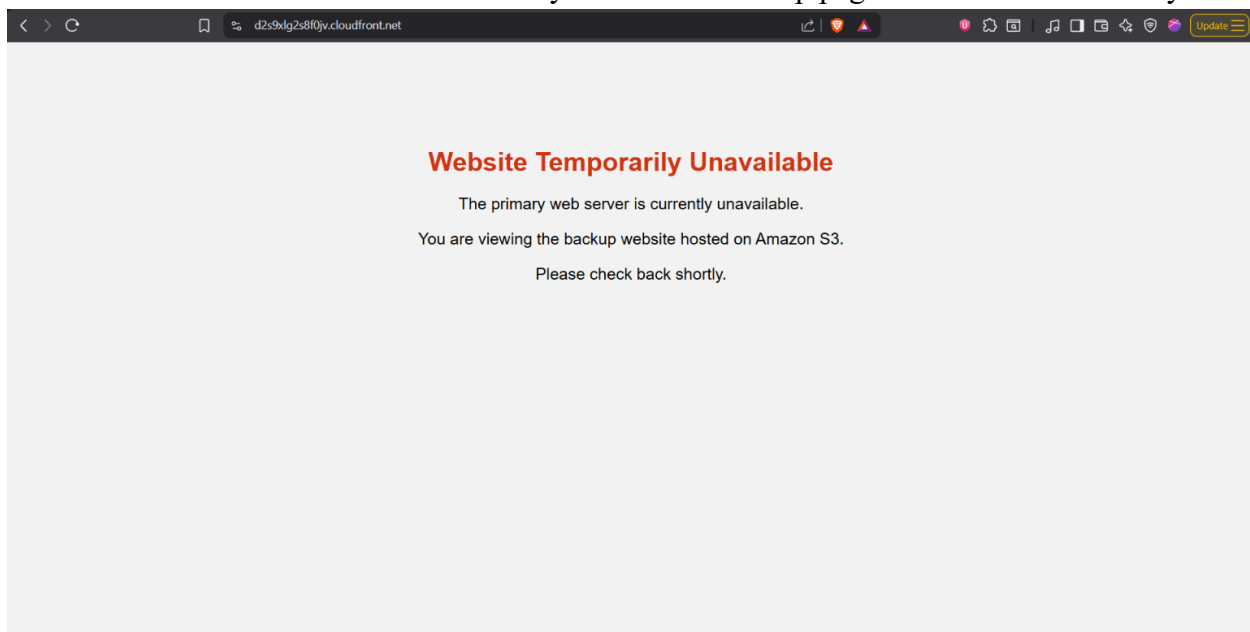
Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0b381782cd6802c0b	IPv4	HTTP	TCP	80	0.0.0.0/0
-	sgr-0b26aac64f08719d8	IPv4	SSH	TCP	22	0.0.0.0/0

Edit inbound rules

- **Remove the HTTP Group to simulate failure.**



- Refresh the CloudFront URL and verify that the S3 backup page is served automatically.



- Restore the EC2 HTTP rule if the primary origin needs to be brought back online.
- Verify that **CloudFront Origin Groups** successfully switched from the primary EC2 origin to the secondary S3 origin during failure.